

AURA — Autonomous Voice Utility & Response Agent

Local-First Autonomous Voice AI with AMD Acceleration

 100% Offline Privacy

 AMD ROCm Acceleration

 Autonomous Tool Engine

Critical Challenges with Cloud Voice AI



Privacy & Confidentiality Risks

Standard cloud voice assistants upload raw microphone audio, personal transcripts, and sensitive document data to third-party remote servers—exposing confidential corporate and personal data.



High Latency & Cloud Dependency

Constant network round-trips introduce unpredictable voice latency. Document search and tool execution fail entirely without an active internet connection.

Meet **AURA** — Autonomous Local Assistant



100% Local Voice & Document Processing

Operates completely on your machine. Voice recognition, multi-step tool execution, and local document RAG run offline without data ever leaving your hardware.



Granular Permission System

Enforces three strict permission tiers: READ, WRITE, and DESTRUCTIVE. Requires explicit user approval before executing state-modifying tools.

AMD ROCm Hardware Acceleration



PyTorch HIP & rocm-smi Integration

Direct inspection of AMD GPU utilization, VRAM allocation, and ROCm driver runtime for fast local tensor computation.

ROCm 6.1.2 | RX 7900 XTX | WMMa Acceleration



Local RAG & Streaming Pipeline

Fast vector similarity search across local PDF/Markdown files connected to a real-time FastAPI WebSocket event pipeline (/ws/voice).

WebSocket /ws/voice | Sub-second E2E Latency

Key Workflows & Demonstration



1. Document RAG Search

Query private project documents via voice and retrieve instant answers backed by verifiable source citations.



2. Task & Action Approval

Identify incomplete tasks, trigger draft summaries, and prompt for interactive UI permission approval.



3. Hardware Diagnostics

Inspect real-time AMD GPU metrics, VRAM usage, driver versions, and throughput stats in the live AMD tab.